

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Chandubhai S Patel Institute of Technology

3rd Semester B.Tech.(EC/ME/CL/EE)

University Exam April-2012

MA-202 Engineering Mathematics-III

Date: 30/04/'12

Time: 01:30 pm to 04:30 pm

Maximum Marks: 70

Instruction:

1. This question paper contains two sections **Section-I** and **Section-II**.
2. **Section-I** carries 35 and **Section-II** carries 35 marks.
3. Section wise separate answer book to be used.
4. Figure to the right indicates the distribution of full marks for the respective question.

Section-IQ.1 (A) Find the Fourier series $f(x) = x^2$ in the interval $-\pi \leq x \leq \pi$. [3](B) Find Laplace transform of function (i) $\sin 2t \cos 3t$ (ii) $\frac{e^{at} - 1}{a}$. [4]Q.2 (A) Find the Laplace transform of the function (i) $t^2 \cos at$ (ii) $t^2 + at + b$. [5](B) Solve $y'' + 3y' + 2y = e^t$ given $y(0) = 1$ and $y'(0) = 0$, using Laplace transform. [5](c) Obtain Fourier series for the function $f(x) = x$ in the range $(0, 2)$. [4]**OR**Q.2 (A) Find the Laplace transform of the function $f(t) = \begin{cases} t, & 0 < t < 2 \\ t-1, & 2 < t < 3 \\ 7, & t > 3 \end{cases}$ [5](B) Using Laplace transform solve $y'' + y = t$ given that $y(0) = 0, y'(0) = 1$. [4](C) Find Laplace inverse of (i) $\frac{s+7}{s^2+2s+2}$ (ii) $\frac{5s}{s^2-25}$. [5]Q.3 (A) Solve $x^3 - 18x - 35 = 0$ by Cardon's Method. [4](B) Find Fourier expression of $f(x) = x + x^2$ for $-\pi < x < \pi$.And deduce that $\frac{\pi^2}{6} = 1 + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \dots$ [5](C) Using Ferrari Method to Solve $x^4 - 12x^3 + 41x^2 - 18x - 72 = 0$. [5]**OR**Q.3 (A) Solve $x^3 - 27x + 57 = 0$ by Cardon's Method. [5](B) Find Fourier series of $f(x)$, where $f(x) = \begin{cases} \pi + x, & -\pi < x < 0; \\ \pi - x, & 0 < x < \pi. \end{cases}$ [5](C) Solve $x^4 - 10x^3 + 35x^2 - 50x + 24 = 0$ Using Ferrari Method. [4]

Section-II

- Q.4 (A) State Divergence theorem. [2]
 (B) An e.m.f $E \sin pt$ is applied at $t=0$ to a circuit containing a condenser of capacity C and inductance L in series. the current i satisfies the equation $L \frac{di}{dt} + \frac{1}{C} \int i dt = E \sin pt$
 If $p^2 = \frac{1}{LC}$ and initially the charge q and the current i are zero. Show that the current at time t is $Et \frac{\sin pt}{2L}$ where $i = \frac{dq}{dt}$. [5]

- Q.5 (A) A particle moves along the curve $x = t^3 + 1, y = t^2, z = 2t + 5$ where t is the time. Find the components of its velocity and acceleration at time $t=1$ in the direction $2\hat{i} + 3\hat{j} + 6\hat{k}$. [4]
 (B) Show that $v(x, y, z) = 2xyz\hat{i} + (x^2z + 2y)\hat{j} + x^2yk$ is irrotational and find a scalar function $u(x, y, z)$ such that $v = \text{grad}(u)$. [5]
 (C) Find the directional derivative of $\text{div}(u)$ at the point $(1, 2, 2)$ in the direction of the outer normal of the sphere $x^2 + y^2 + z^2 = 9$ for $u = x^4\hat{i} + y^4\hat{j} + z^4\hat{k}$. [5]

OR

- Q.5 (A) Find a unit vector normal to the surface $x^2 + 3y^2 + 2z^2 = 6$ at $P(2, 0, 1)$. [4]
 (B) If $f = xy^2\hat{i} + 2x^2yz\hat{j} - 3yz^2\hat{k}$, then find $\text{div } f$ and $\text{curl } f$ at the point $(1, -1, 1)$. Is f is Solenoidal? [5]
 (C) (i) Prove that curl of a constant vector is null vector.
 (ii) If $r^2 = x^2 + y^2 + z^2$, find $\text{grad } r^n$. [5]

- Q.6 (A) Evaluate $\int_c F \cdot dr$ where $f = (x^2 - y^2)\hat{i} + xy\hat{j}$ and c is the arc of $y = x^3$ from $(0, 0)$ to $(2, 8)$. [4]
 (B) State Green's theorem and evaluate $\int_c e^{-x} \sin y \, dx + e^{-x} \cos y \, dy$ where c is rectangle with vertices $(0, 0), (\pi, 0), (\pi, \pi/2), (0, \pi/2)$. [5]
 (C) Verify Stoke's theorem when $f = y\hat{i} + z\hat{j} + x\hat{k}$ and surface s is the part of the sphere $x^2 + y^2 + z^2 = 1$ above the xy -plane. [5]

OR

- Q.6 (A) Apply the divergence theorem to compute $\iint_S \bar{u} \cdot \hat{n} \, ds$, where S is the surface of the cylinder $x^2 + y^2 = 9$ bounded by the planes $z = 0, z = b$ where $\bar{u} = \hat{i}x - \hat{j}y + \hat{k}$. [5]
 (B) A vector field F is given by $f = \sin y \hat{i} + x(1 + \cos y)\hat{j}$ evaluate the line integral $\int_c f \cdot dr$ where c is the circular path given by $x^2 + y^2 = a^2$ Using Green's theorem. [5]
 (C) Evaluate $\int_c f \cdot dr$ where $f = xy\hat{i} + yz\hat{j} + zx\hat{k}$ and c is a curve $r = t\hat{i} + t^2\hat{j} + t^3\hat{k}$ where t varies from -1 to 1 . [4]

